

Polycystic Ovary Syndrome (PCOS) Analysis and Predictive Modeling

Data Preprocessing

Several preprocessing steps were performed before modeling:

Feature Engineering

- Construction of Body Mass Index (BMI)-related variables.
- Creation of a binary abortion-history indicator.
- Creation of a new variable representing the total number of ovarian follicles.

Feature Selection

The following variables were removed due to redundancy or high correlation with derived variables:

- Weight
- Height
- FSH
- LH
- Waist circumference
- Hip circumference
- Left follicle count
- Right follicle count

Outlier Detection

Medical literature was consulted to validate extreme observations.

The following values were considered measurement errors and removed:

- $FSH/LH = 1372.826$
- Vitamin D3 = 6014.86
- Vitamin D3 = 5418.60

Missing Values

A small number of missing observations were removed using complete-case analysis.

Exploratory Multivariate Analysis

Several unsupervised learning techniques were applied to better understand the structure of the data.

Correlation Analysis

- Pearson correlation matrix
- Identification of relationships among clinical variables

Principal Component Analysis (PCA)

Used to:

- Detect multicollinearity
- Reduce dimensionality
- Identify the most influential clinical measurements

Factor Analysis (FA)

Performed to:

- Identify latent clinical factors
- Summarize correlated variables into interpretable components

Multidimensional Scaling (MDS)

Distance-based representations were constructed using:

- Euclidean distance
- Manhattan distance
- Gower distance

to visualize similarities among patients.

Multiple Correspondence Analysis (MCA)

Applied to categorical variables in order to:

- Explore associations among categories
- Detect unusual observations
- Visualize patient profiles

Predictive Modeling

Several supervised classification techniques were evaluated to predict PCOS status.

Linear Discriminant Analysis (LDA)

Assumptions evaluated:

- Multivariate normality
- Homogeneity of covariance matrices

Tree-Based Models

Decision tree methods were implemented to generate interpretable diagnostic rules based on clinical variables.

Random Forest

Random Forest models were trained to:

- Improve predictive performance
 - Estimate variable importance
 - Reduce overfitting
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Technologies and Libraries

This project was developed in **R** using the following packages:

- FactoMineR
 - factoextra
 - ggplot2
 - dplyr
 - caret
 - pROC
 - glmnet
 - psych
 - randomForest
 - rpart
 - rpart.plot
 - MVN
 - biotools
 - cluster
 - ca
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Main Objectives

1. Understand the multivariate structure of PCOS-related clinical variables.

2. Identify important risk factors associated with the syndrome.
 3. Compare exploratory and predictive statistical techniques.
 4. Develop interpretable classification models for PCOS diagnosis.
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